

Combinations

List all possible combinations.

1) T, V, W, taken two at a time

TV VW
TW

2) A, B, C, D, taken two at a time

AB BC
AC BD
AD CD

3) ☺, ☀, ♥, ▲, taken three at a time

☺☀♥ ☀♥▲
☺☀▲
☺♥▲

4) 4, 5, 6, 7, taken four at a time

4567

Evaluate each expression.

5) ${}_{22}C_{20}$

231

6) ${}_{11}C_8$

165

7) ${}_{12}C_8$

495

8) ${}_{25}C_{23}$

300

9) ${}_{24}C_5$

42,504

10) ${}_{17}C_{10}$

19,448

11) $4 \cdot {}_{18}C_{11}$

127,296

12) ${}_{20}C_{16} + 1$

4,846

13) $\frac{{}_{20}C_5}{8}$

1,938

14) $-6 + {}_{19}C_5$

11,622

Critical thinking questions:15) Explain why ${}_nC_2 = {}_nC_{n-2}$ Choosing $n - 2$ means two are being left behind.
You could think of it choosing those two.

16) Write a combination that equals 12345

 ${}_{12345}C_1$